MINISTRY OF COMMUNICATIONS AND INFORMATION TECHNOLOGY (MCIT)
DIGITAL EGYPT PIONEERS INITIATIVE (DEPI)
DATA ANALYSIS TRACK – PHASE 1 CAPSTONE PROJECT PROPOSAL

PROJECT CHARTER & SPECIFICATION DOCUMENT

1. Project Information

Project Title	MTA Daily Ridership: Post-COVID Recovery Analysis & Next-Month Forecast
Track & Phase	DEPI – Data Analysis Track – Phase 1
Submission Date	October 16, 2026 (Official Deadline)
GitHub Repository	**https://github.com/Salem-Emad/MTA-Daily-Ridership-Data-Analysis**
Team Leader & Repo Administrator	Salem Emad Salem
Team Members & Roles	• Salem Emad Salem – Team Leader & Repository Administrator; Project Proposal • Dalia El-Barbary – Gantt Chart, Task Distribution & Risk Plan • Rana Osama – KPIs & Success Measures • Mohamed Hossam – Requirements Specification (SRS) • Mira Morris – Team Member (joined after the Phase 1 task split) • Phase 2–4 tasks are selected by the members and tracked in the project Gantt Chart.

2. Project Overview

The Metropolitan Transportation Authority (MTA) is North America's largest transportation network, serving over 15 million people across the New York metropolitan area through subways, buses, commuter railroads, paratransit, and vehicular crossings. This analytical project examines public transit recovery across the New York City region following the catastrophic operational disruption caused by the COVID-19 pandemic. Utilizing official daily ridership data from March 1, 2020, to October 31, 2024, the project evaluates recovery velocities across seven distinct transportation modes against pre-pandemic baselines. Through advanced data modeling in Microsoft Excel and Power BI, our objective is to deliver actionable, data-driven intelligence to transit authorities, urban planners, and municipal decision-makers to optimize scheduling, budgetary allocations, and long-term fleet management.

3. Problem Statement

Since March 2020, daily passenger transit behavior across New York City has experienced lasting disruptions, likely linked to hybrid work patterns, demographic shifts, and evolving travel habits. MTA executive management and municipal transit authorities lack a consolidated, standardized analytical framework to quantitatively compare recovery rates across all transit modes against pre-pandemic volumes. Furthermore, transit operations face critical uncertainties regarding weekly demand fluctuations, recurring monthly seasonality, and predictive ridership trajectories. These blind spots directly affect transit operators, fiscal budget committees, daily commuters, and municipal planning departments, leading to fleet misallocation, inaccurate subsidy forecasts, and sub-optimal service schedules.

4. Measurable Project Objectives (SMART)

• **Objective 1: Multi-Modal Recovery Benchmarking –** Analyze and benchmark the post-pandemic recovery trajectory of all 7 MTA transit modes across 1,706 operational days (March 1, 2020 – October 31, 2024), measuring the exact percentage of comparable pre-pandemic volume per mode.

• **Objective 2: Seasonality & Peak Pattern Identification –** Uncover cyclical ridership behavior by identifying the top 3 peak days and the seasonal trough months, comparing weekdays vs. weekends, with mode-level detail.

• **Objective 3: Predictive Ridership Forecasting –** Formulate an empirical statistical forecast model for total passenger ridership (strictly aggregating mass-transit passenger modes: Subways, Buses, LIRR, Metro-North, and SIR; isolating vehicular crossings and paratransit) across November 2024, achieving a Mean Absolute Percentage Error (MAPE) ≤ 10% verified against historical back-testing splits.

• **Objective 4: Interactive Business Intelligence Dashboard –** Design and deploy an executive-ready, interactive Power BI analytical dashboard consisting of at least 4 dedicated analytical views (executive overview, recovery analysis, patterns and seasonality, and forecast).

• **Objective 5: Version Control & Documentation –** Maintain professional version control and end-to-end documentation on GitHub, featuring an exhaustive README, Power Query transformation scripts, data dictionaries, and documented test cases.

5. Project Scope: In-Scope vs. Out-of-Scope

To ensure structured project execution within the Phase 1 timeframe, clear project boundaries have been formally established:

In-Scope Elements	Out-of-Scope Elements
• Analysis of the provided official daily dataset (1,706 days: March 1, 2020 – October 31, 2024). • All 7 MTA transport modes (Subway, Bus, LIRR, Metro-North, Access-A-Ride, Bridges & Tunnels, Staten Island Railway). • End-to-end data cleansing, outlier remediation, and KPI calculations. • Passenger ridership forecast for November 2024 with MAPE backtest (Target: MAPE ≤ 10%). • Multi-page interactive Power BI dashboard (at least 4 pages) featuring dynamic slicers, Star Schema, and DAX measures.	• Real-time or live API stream ingestion (static historical analysis only). • External environmental and demographic datasets (weather conditions, public event feeds, local COVID infection tallies). • Micro-level station-by-station, train car-level, or individual bus route spatial modeling. • Production-level cloud deployment, automated orchestration pipelines, or enterprise streaming architectures. • Ticket fare evasion metrics, revenue audit trails, or individual passenger demographic profiling.

6. Dataset Description & Data Quality Audit

Source: MTA Daily Ridership Data: Beginning 2020, published by the MTA on NY Open Data (https://data.ny.gov/d/vxuj-8kew). The portal now titles this dataset "MTA Daily Ridership Data: 2020 - 2025" and marks it as deprecated; next-day estimates without the pre-pandemic comparison are published in a separate dataset. This project uses the downloaded snapshot described below.

The project dataset comprises 1,706 complete calendar days spanning from March 1, 2020, to October 31, 2024, without missing observation dates. The schema contains 15 fields consisting of a primary temporal key ('Date') and paired metrics (Total Volume and Percentage of Comparable Pre-Pandemic Day) across seven transit systems:

Transit System	Volume Metric	% Pre-Pandemic Field	Key Observations & Quality Nuances
Subways	Total Estimated Ridership	Subways: % of Comparable Pre-Pandemic Day	Primary rapid transit system; largest passenger volume in NYC.
Buses	Total Estimated Ridership	Buses: % of Comparable Pre-Pandemic Day	Covers local and express routes across all 5 boroughs.
Long Island Rail Road (LIRR)	Total Estimated Ridership	LIRR: % of Comparable Pre-Pandemic Day	Commuter rail connecting suburban Long Island to Manhattan.
Metro-North Railroad	Total Estimated Ridership	Metro-North: % of Comparable Pre-Pandemic Day	Commuter rail connecting northern suburbs & CT to Grand Central.
Access-A-Ride	Total Scheduled Trips	Access-A-Ride: % of Comparable Pre-Pandemic Day	Paratransit service measuring scheduled trips rather than riders.
Bridges and Tunnels	Total Traffic	Bridges and Tunnels: % of Comparable Pre-Pandemic Day	Measures motor vehicle crossings, NOT passenger headcounts.
Staten Island Railway (SIR)	Total Estimated Ridership	Staten Island Railway: % of Comparable Pre-Pandemic Day	Exhibits 32 zero-ridership records (21 June 2020 – 15 August 2021) and 54 zero values in its % field.

Critical Data Quality Considerations Identified:

1. Staten Island Railway Zero Values: The ridership field has 32 zero values between 21 June 2020 and 15 August 2021, and the percentage field has 54 zero values between 30 May 2020 and 30 October 2022. The dataset does not document the cause, so these zeros are treated as probable missing observations rather than confirmed zero ridership. In the Power Query / ETL phase they will be reviewed and, where they represent missing observations, converted to nulls, preventing artificial downward bias in historical averages (aligned with User Story US-07).

2. Disparate Measurement Units: The dataset contains heterogeneous metrics: passenger counts (Subways, Buses, LIRR, Metro-North, SIR), vehicular traffic counts (Bridges & Tunnels), and scheduled trips (Access-A-Ride). Direct summation across these units is not meaningful, so total passenger ridership includes only Subways, Buses, LIRR, Metro-North, and SIR, while Bridges & Tunnels and Access-A-Ride are reported separately.

3. Pre-Pandemic Baselines: Percentages are pre-calculated by the MTA against an equivalent (comparable) pre-pandemic day, providing standardized benchmarking across modes.

4. Estimation Methods: Ridership figures are estimates (for example, LIRR and Metro-North are modeled from ticket sales, and Access-A-Ride counts scheduled rather than completed trips), and recent figures may be revised by the MTA.

7. Project Lifecycle Deliverables & Milestone Schedule

The project adheres to the official four-phase lifecycle mandated by DEPI, ensuring end-to-end analytical rigor, architectural governance, and milestone compliance:

Phase & Focus Area	Official Deadline	Core Deliverables	Target Technology & Scope
Phase 1: Planning & Requirements	**16 October 2026 (Official Deadline)**	• Project Charter & Proposal • Gantt Schedule & Risk Plan • KPI Definitions & Framework • SRS Requirements Specification	Word, Excel, PDF, GitHub Repo Setup
Phase 2: Design & Analysis	**06 November 2026**	• Final Problem Statement & Objectives • Use Case Diagram & ERD (Star Schema design) • System DFD • Dashboard Wireframes • Tech Stack	Draw.io, PowerPoint / Figma, Word
Phase 3: Implementation & Forecasting	**30 November 2026**	• Power Query ETL & Star Schema Model • Multi-page Dashboard (≥ 4 pages) • Advanced DAX Measures & KPIs • November 2024 Ridership Forecast • Model Validation (MAPE ≤ 10%) • Clean GitHub Code & README	Power BI Desktop, Power Query, DAX, GitHub
Phase 4: Testing & Final Delivery	**04 December 2026**	• Test Cases, Bug Reports & RTM Validation • User Guide & Technical Documentation • Final Presentation & Infographic • Demo Video (optional)	GitHub, PowerPoint, Word / PDF

8. Related Work & Literature Review

1. Official Data Source – NY Open Data, MTA Daily Ridership Data (https://data.ny.gov/d/vxuj-8kew): The MTA publishes the daily dataset used in this project and compares each day with a comparable pre-pandemic day. It provides raw measurements only: it does not include a consolidated multi-mode analysis, a seasonality view, or a forecast. Our project adds these analytical layers.

2. Public-Sector Analyses – NYC Comptroller, "Riders Return" (https://comptroller.nyc.gov/reports/riders-return), and NY State Comptroller, MTA fiscal recovery report 4-2026 (https://www.osc.ny.gov/files/reports/pdf/report-4-2026.pdf): These reports describe uneven recovery across modes: subway and bus recovery have followed different paths, a weekday–weekend gap persists, and commuter rail and paratransit have recovered faster than subway and bus. They give context for our mode-by-mode comparison. Our project reproduces this comparison interactively from the daily data and adds a next-month forecast.

3. Academic and Community Work – Princeton senior thesis on remote work and NYC transit (https://theses-dissertations.princeton.edu/handle/88435/dsp01j3860b387) and Maven Analytics "MTA Post-Pandemic Recovery Dashboard" (https://mavenanalytics.io/project/22211): The thesis studies how remote-work prevalence relates to transit use across NYC-area systems, and the community dashboard

presents recovery by mode for the same MTA data. Our project does not assume a particular finding; it adds a documented star-schema model, a next-month forecast with a measured error (MAPE), and complete test and documentation artifacts.

9. Expected Outcomes & Business Impact

• **Comprehensive Diagnostic Assessment:** Clear visibility into which transit modes have reached their pre-pandemic level and which have not. Preliminary figures from the October 2024 data (average % of comparable pre-pandemic day): Access-A-Ride 127%, Bridges and Tunnels 103%, LIRR 96%, Metro-North 86%, Subways 75%, Buses 66%, and Staten Island Railway 53%.

• **Fleet & Budget Optimization Insights:** Actionable intelligence for transit authorities to dynamically align peak schedule capacity with documented weekly and seasonal commuter demand surges.

• **Forecast with Measured Accuracy:** A monthly forecast for November 2024 with a measured backtest error (MAPE; target ≤ 10%), supporting procurement, energy planning, and municipal budget allocation.

• **Complete BI Deliverables:** A multi-page (≥ 4 pages) interactive Power BI dashboard accompanied by complete GitHub documentation that meets the DEPI deliverable requirements.